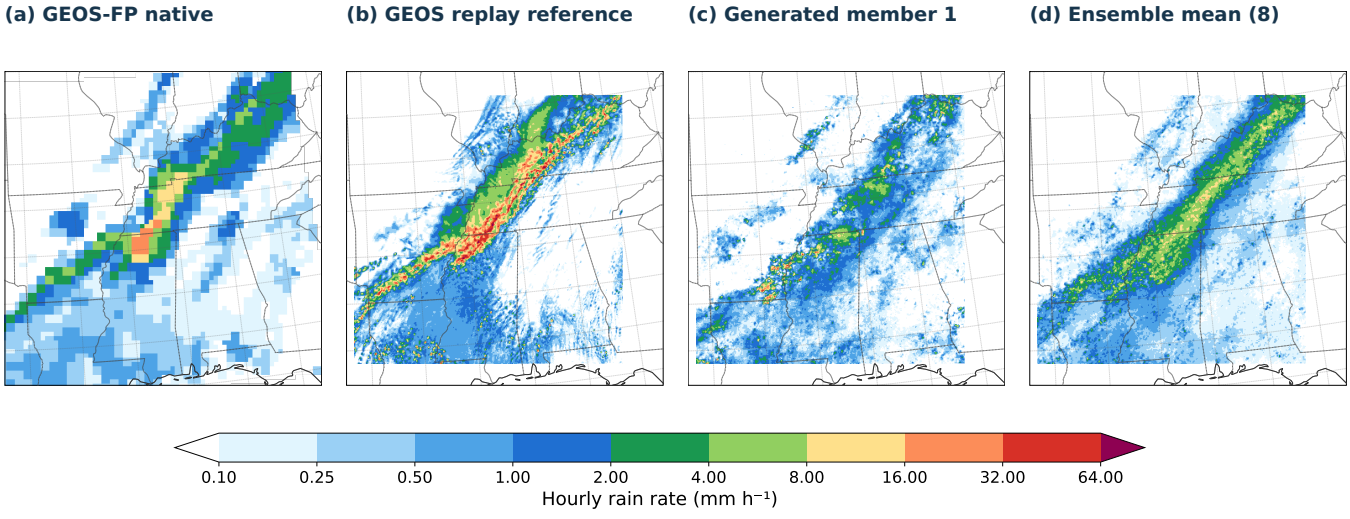


Actual precipitation | 2026-03-07 19:30:00 UTC | epoch 243, EMA, 64 Heun steps, spectral

1. CONDITIONS

Current + previous-hour GEOS-FP
Local and broad spatial context
Static surface and time features

2. HYBRID FLOW MATCHING

Direct square-root precipitation
Residual temperature, pressure, winds
Coarse-residual humidity

3. ENSEMBLE SAMPLING

Shared-state tiled Heun solver
Independent member noise draws
Optional paired spectral adjustment

Area/channel-weighted velocity loss; 192^2 patches include halo supervision.

GEOS replay: dynamical reference on the nominal 2–3 km grid. Predictions: learned downscaling.